Contracts to Effects Compiler Correctness

(Local Handler)

In this part, we provide the proof for a compiler version using local handler. Instead of a global handler outside, a flat contract is compiled into an effect and handle it on the spot. All other compilation rules stay the same. Semantics of handlers can be either Deep or Shallow. However, because Deep Handlers push a new (unused) handler each time a check is performed, Shallow Handlers are chosen for this compiler implementation.

$$\frac{\Gamma \vdash e_1 : \tau \to \mathsf{bool} \rightarrowtail^l e_3 \qquad \Gamma \vdash e_2 : \tau \rightarrowtail^l e_4}{\Gamma \vdash \mathsf{mon}_j^{k,l}(\mathsf{flat}(e_1), e_2) : \tau \rightarrowtail^l \mathtt{handle}_s(\mathtt{h}_{\mathscr{C}}, (\mathtt{perform}\ \mathscr{C}\ \tau\ \langle k, \langle e_3, e_4\rangle\rangle))}\ \text{Compile-Flat}$$

$$\begin{aligned} \mathtt{h}_{\mathscr{C}} : \mathtt{lab} &= \{\mathscr{C} \to \forall\alpha.\langle \mathtt{blab}, \langle \alpha \to \mathsf{bool}, \alpha\rangle\rangle \to \alpha\} \\ &= \{\mathscr{C} \to \lambda\langle l, \langle e, v\rangle\rangle.\lambda k.\ \text{if } e\ v \text{ then } k\ v \text{ else } \mathtt{error}_l\} \\ &\text{where } \lambda\langle l, \langle e, v\rangle\rangle \text{ deconstructs the argument as tuple pattern} \end{aligned}$$

Relations

Before we define relations that will be used in the simulation proof below. We define:

- Context E^* is any context without $\mathsf{check}_j^k(e, v)$ in Contracts.
- Context $E^+ = \square \mid E^+\left[E^*\left[\mathsf{check}_j^k(\square, v)\right]\right]$

$\boxed{R_v \subseteq \mathsf{Contracts}\ v \times \mathtt{Effects}\ v}$

Abuse of notation: If $(v, v') \in R_v$, we write $v \sim_{R_v} v'$ or $v \sim v'$. We write $\tilde{v}$ for any value $\tilde{v}$ satisfying $v \sim \tilde{v}$.

$$\begin{aligned} x &\sim x \\ b &\sim b \\ N &\sim N \\ \lambda x : \tau.e &\sim \lambda x : \tau^*.\tilde{e} \\ \mathsf{guard}_j^{k,l}(\kappa_1 \to \kappa_2, v) &\sim \tilde{e} \\ & \qquad e = \lambda x : \tau.\mathsf{mon}_j^{k,l}\left(\kappa_2, v\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\right) \end{aligned}$$

$$\begin{array}{rcl}
\mathsf{guard}_j^{k,l}(\kappa_1 \to_i \lambda y.\kappa_2, v) & \sim & \tilde{e} \\
& & e = \lambda x : \tau.\mathsf{mon}_j^{k,l}\Big(\kappa_3, v\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\Big) \\
& & \kappa_3 = \kappa_2\Big[y := \mathsf{mon}_j^{l,j}(\kappa_1, x)\Big]
\end{array}$$

$\boxed{R_e \subseteq \mathsf{Contracts}\ e \times \mathtt{Effects}\ e}$

Abuse of notation: If $(e, e') \in R_e$, we write $e \sim_{R_e} e'$ or $e \sim e'$. We write $\tilde{e}$ for any expression $\tilde{e}$ satisfying $e \sim \tilde{e}$.

$$\begin{array}{rcl}
v & \sim & \tilde{v} \\
e_1\ e_2 & \sim & \tilde{e}_1\ \tilde{e}_2 \\
\mathsf{mon}_j^{k,l}(\mathsf{flat}(e_1), e_2) & \sim & \mathtt{handle}_s(\mathtt{h}_{\mathscr{C}}, \mathtt{perform}\ \mathscr{C}\ \langle k, \langle \tilde{e}_1, \tilde{e}_2 \rangle\rangle) \\
\mathsf{mon}_j^{k,l}(\kappa_1 \to \kappa_2, e) & \sim & \tilde{e}_1\ \tilde{e} \\
& & e_1 = \lambda f.\lambda x.\mathsf{mon}_j^{k,l}\Big(\kappa_2, f\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\Big) \\
\mathsf{mon}_j^{k,l}(\kappa_1 \to_i \lambda y.\kappa_2, e) & \sim & \tilde{e}_1\ \tilde{e} \\
& & e_1 = \lambda f.\lambda x.\mathsf{mon}_j^{k,l}\Big(\kappa_3, f\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\Big) \\
& & \kappa_3 = \kappa_2\Big[y := \mathsf{mon}_j^{l,j}(\kappa_1, x)\Big] \\
\mathsf{blame}_j^k & \sim & \mathtt{error}_k
\end{array}$$

$\boxed{R_{E^*} \subseteq \mathsf{Contracts}\ E^* \times \mathtt{Effects}\ E^*}$

Abuse of notation: If $(E^*, E'^*) \in R_{E^*}$, we write $E^* \sim_{R_{E^*}} E'^*$ or $E^* \sim E'^*$. We write $\tilde{E}^*$ for any expression $\tilde{E}^*$ satisfying $E^* \sim \tilde{E}^*$.

$$\begin{array}{rcl}
\square & \sim & \square \\
E^*\ e & \sim & \tilde{E}^*\ \tilde{e} \\
v\ E^* & \sim & \tilde{v}\ \tilde{E}^* \\
\mathsf{mon}_j^{k,l}(\mathsf{flat}(e), E^*) & \sim & \mathtt{handle}_s\big(\mathtt{h}_{\mathscr{C}}, \mathtt{perform}\ \mathscr{C}\ \langle k, \langle \tilde{e}_1, \tilde{E}^* \rangle\rangle\big) \\
\mathsf{mon}_j^{k,l}(\kappa_1 \to \kappa_2, E^*) & \sim & \tilde{e}_1\ \tilde{E}^* \\
& & e_1 = \lambda f.\lambda x.\mathsf{mon}_j^{k,l}\Big(\kappa_2, f\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\Big) \\
\mathsf{mon}_j^{k,l}(\kappa_1 \to_i \lambda y.\kappa_2, E^*) & \sim & \tilde{e}_1\ \tilde{E}^* \\
& & e_1 = \lambda f.\lambda x.\mathsf{mon}_j^{k,l}\Big(\kappa_3, f\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\Big) \\
& & \kappa_3 = \kappa_2\Big[y := \mathsf{mon}_j^{l,j}(\kappa_1, x)\Big]
\end{array}$$

$R_{E^+} \subseteq \mathsf{Contracts}\ E^+ \times \mathtt{Effects}\ E^+$

Abuse of notation: If $(E^+, E'^+) \in R_{E^+}$, we write $E^+ \sim_{R_{E^+}} E'^+$ or $E^+ \sim E'^+$. We write $\tilde{E}^+$ for any expression $\tilde{E}^+$ satisfying $E^+ \sim \tilde{E}^+$.

$$\begin{array}{lcl} & \Box \sim \Box & \\ E^+\left[E^*\left[\mathsf{check}^k_j(\Box, v)\right]\right] & \sim & \text{if } \Box \text{ then } \left(\lambda x.\tilde{E}^+\left[\tilde{E}^*[x]\right]\right) \tilde{v} \text{ else } \mathtt{error}_k \end{array}$$

Lemma 0.1. (Relation is closed)

If $E^* \sim \tilde{E}^*$, $e \sim \tilde{e}$ and e has no check in evaluation position, then $E^*[e] \sim \tilde{E}^*[\tilde{e}]$.

Proof. By induction on $E^* \sim \tilde{E}^*$. Follows similarly to [?] because the relations (R_e, R_E) are the same. □

Lemma 0.2. (Relation is closed under substitution)

If $e \sim \tilde{e}$ and $v \sim \tilde{v}$ then $e[x := v] \sim \tilde{e}[x := \tilde{v}]$.

Proof. By induction on $e \sim e'$. Follows [?] because relations (R_v, R_e) are the same. For this compiler, there is a single case that has their relation different from Deep/Shallow compilers.

- Case $\mathsf{mon}^{k,l}_j(\mathsf{flat}(e_1), e_2) \sim \mathtt{handle}_s(\mathtt{h}_{\mathscr{C}}, \mathtt{perform}\ \mathscr{C}\ \langle k, \langle \tilde{e}_1, \tilde{e}_2 \rangle\rangle)$

  $e_1[x := v] \sim \tilde{e}_1[x := v']$ I.H.
  $e_2[x := v] \sim \tilde{e}_2[x := v']$ I.H.
  $\mathsf{mon}^{k,l}_j(\mathsf{flat}(e_1), e_2)[x := v] \sim \mathtt{handle}_s(\mathtt{h}_{\mathscr{C}}, \mathtt{perform}\ \mathscr{C}\ \langle k, \langle \tilde{e}_1, \tilde{e}_2 \rangle\rangle)[x := v']$ Def.

- Other cases follow [?] because their relations (R_e) are the same.

□

Lemma 0.3. (Relation decomposition through check-free contexts)

If $E^*[e] \sim e'$, then there exist E'^* and e'_0 such that $E^* \sim E'^*$ and $e \sim e'_0$ and $e' = E'^*[e'_0]$.

Proof.

By induction on the check-free context E^*. Follows [?] because relations (R_E, R_e) are the same. □

Lemma 0.4. (Breaking down Context E)

For all contexts E, uniquely decompose $E = E^+[E^*]$.

Uniqueness observation: Recall that E^* has no check contexts.

- If E does not have check, then it is $E^* = E$, and $E^+ = \square$.
- Else there is only a single decomposition to get out E^* without check contexts

Proof. By induction on the structure of E. Follows [?] because the contexts E, E^+, E^* are the same. □

Lemma 0.5. (Compiler is subset of Relation)

If $\Gamma \vdash e : \tau \rightarrowtail^l e'$ then $e \sim e'$

Proof.

By induction on structure of $\Gamma \vdash e_1 : \tau \rightarrowtail^l e_2$.

- Case

$$\frac{\Gamma \vdash e_3 : \tau \to \mathsf{bool} \rightarrowtail^l e_5 \qquad \Gamma \vdash e_4 : \tau \rightarrowtail^l e_6}{\Gamma \vdash \mathsf{mon}_j^{k,l}(\mathsf{flat}(e_3), e_4) : \tau \rightarrowtail^l \mathtt{handle}_s(\mathtt{h}_{\mathscr{C}}, \mathtt{perform}\ \mathscr{C}\ \tau\ \langle k, \langle e_5, e_6\rangle\rangle)}$$

$e_2^* = \mathtt{handle}_s(\mathtt{h}_{\mathscr{C}}, \mathtt{perform}\ \mathscr{C}\ \langle k, \langle e_5^*, e_6^*\rangle\rangle)$

$e_3 \sim e_5^*$ I.H.

$e_4 \sim e_6^*$ I.H.

$e_1 \sim e_2^*$

- Case

$$\frac{\Gamma \vdash e_3 : \tau_1 \to \tau_2 \rightarrowtail^l e_4 \qquad \Gamma \vdash \lambda f.\lambda x.\mathsf{mon}_j^{k,l}\left(\kappa_2, f\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\right) : (\tau_1 \to \tau_2) \to (\tau_1 \to \tau_2) \rightarrowtail^l e_5}{\Gamma \vdash \mathsf{mon}_j^{k,l}(\kappa_1 \to \kappa_2, e_3) : \tau_1 \to \tau_2 \rightarrowtail^l e_5\ e_4}$$

$e_2^* = e_5^*\ e_4^*$

$e_3 \sim e_4^*$ I.H.

$\lambda f.\lambda x.\mathsf{mon}_j^{k,l}\left(\kappa_2, f\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\right) \sim e_5^*$ I.H.

$e_1 \sim e_2^*$

- Case

$$\frac{\Gamma \vdash e_3 : \tau_1 \to \tau_2 \rightarrowtail^l e_4 \qquad \Gamma \vdash \lambda f.\lambda x.\left(\lambda y.\mathsf{mon}_j^{k,l}\left(\kappa_2, f\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\right)\right)\ \mathsf{mon}_j^{l,j}(\kappa_1, x) : (\tau_1 \to \tau_2) \to (\tau_1 \to \tau_2) \rightarrowtail^l e_5}{\Gamma \vdash \mathsf{mon}_j^{k,l}(\kappa_1 \to_i \lambda y.\kappa_2, e_3) : \tau_1 \to \tau_2 \rightarrowtail^l e_5\ e_4}$$

$e_2^* = e_5^*\ e_4^*$

$e_3 \sim e_4^*$ I.H.

$\lambda f.\lambda x.\left(\lambda y.\mathsf{mon}_j^{k,l}\left(\kappa_2, f\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\right)\right)\ \mathsf{mon}_j^{l,j}(\kappa_1, x) \sim e_5^*$ I.H.

$e_1 \sim e_2^*$

- Other cases follow similarly, by I.H.

□

Correctness Theorem

$$\mathrm{EQ}(e, e') = \begin{cases} \text{true if } e \sim e' \text{ and } e \neq E^*\left[\mathsf{check}_{j_1}^{k_1}(e_1, v_1)\right] \\ \text{true if } e = E^+\left[E^*\left[\mathsf{check}_{j_1}^{k_1}(e_1, v_1)\right]\right] \text{ and} \\ \qquad e' = \text{if } \tilde{e}_1 \text{ then } \left(\lambda x.\tilde{E}^+\left[\tilde{E}^*[x]\right]\right)\tilde{v}_1 \text{ else } \mathtt{error}_{k_1} \\ \text{true if } e = E\left[\mathsf{blame}_j^k\right] \text{ and } e' = \mathtt{error}_k \\ \text{false otherwise} \end{cases}$$

Theorem 0.6.

If $e \twoheadrightarrow^* b$ and $\cdot \vdash e : \tau \rightarrowtail^l e'$ then $e' \Longrightarrow^* b$

Runs without a wrapper because the local handler resolves the contract effect in-place.

Proof.

$e \sim e'$	Lemma 0.5
$\mathrm{EQ}(e, e')$	Case 1 EQ
$e' \Longrightarrow^* \tilde{b} = b$	Lemma 0.7

□

Lemma 0.7.

If $\mathrm{EQ}(e, e')$ and $e \twoheadrightarrow^* r$ then $e' \Longrightarrow^* \tilde{r}$.

Where $r = v \mid \mathsf{blame}_j^k$ and $\tilde{r} = \tilde{v} \mid \mathtt{error}_k$

Proof.

By induction on the structure of $e \twoheadrightarrow^* r$

- Case $v \twoheadrightarrow^* v$

$\mathrm{EQ}(e, e')$	Given
$e \sim e'$	Case 1 EQ, otherwise invalid
$e' = \tilde{v}$	Inversion
$e' \Longrightarrow^* \tilde{v}$	Reflexivity

- Case $\mathsf{blame}_j^k \twoheadrightarrow^* \mathsf{blame}_j^k$

$\mathrm{EQ}(e, e')$	Given
$e' = \mathtt{error}_k$	Case 3 EQ

$e' \Longrightarrow^* \texttt{error}_k$ Reflexivity

- Case $e \to e_1$ and $e_1 \to^* r$

  $e' \Longrightarrow^* e'_1$ Lemma 0.8
  $\mathrm{EQ}(e_1, e'_1)$ Lemma 0.8
  $e'_1 \Longrightarrow^* \tilde{r}$ I.H.
  $e' \Longrightarrow^* \tilde{r}$ Step

  □

Lemma 0.8.

If $\mathrm{EQ}(e_1, e'_1)$ and $e_1 \to e_2$ then exists e'_2 that $e'_1 \Longrightarrow^* e'_2$ and $\mathrm{EQ}(e_2, e'_2)$

Proof.

By case analysis on $e_1 \to e_2$.

- $E\left[\mathsf{blame}^k_j\right] \to \mathsf{blame}^k_j$, and $E \neq \square$

  $e'_1 = \texttt{error}_k$ Case 3 EQ
  $e'_1 \Longrightarrow^* \texttt{error}_k$ Step and Reflexivity
  $e'_2 = \texttt{error}_k$ Choose

- $E[e_3] \to E[e_4]$ where $e_3 \dashrightarrow e_4$ is a redex

  Proof Sketch: $E = E^+\left[E^*\left[\mathsf{check}^k_j(\square, v)\right]\right]$, then casing on E^+ structure. At the same time, EQ predicate will allow one case analysis $(e_1 ? = E'\left[\mathsf{check}^{k'}_{j'}(e', v')\right])$. If $E^+ = \square$, use Lemma 0.3 to find equivalent $\tilde{E}^*$, else, by EQ case 2 identify the shape of e'_1 by E^+ relation.

  ▸ Case $e_3 = \mathsf{mon}^{k,l}_j(\mathsf{flat}(e), v) \dashrightarrow \mathsf{check}^k_j(e\ v, v) = e_4$

    Actually need $e \neq E'\left[\mathsf{check}^{k'}_{j'}(e', v')\right]$ to obtain $\tilde{e}$. But the proof goes through without stating this fact because we inverse from EQ predicate. When running, this e is never a check, because we don't evaluate inside flat contract.

    $e_1 = E\left[\mathsf{mon}^{k,l}_j(\mathsf{flat}(e), v)\right] = E^+\left[E^*\left[\mathsf{mon}^{k,l}_j(\mathsf{flat}(e), v)\right]\right]$ Lemma 0.4

    – Case $E^+ = \square$

      $e_1 = E^*\left[\mathsf{mon}^{k,l}_j(\mathsf{flat}(e), v)\right]$
      $E^* \sim \tilde{E}^*$ Lemma 0.3
      $e_3 \sim \tilde{e}_3$ Lemma 0.3
      $\tilde{e}_3 = \texttt{handle}_s(\mathtt{h}_{\mathscr{C}}, \texttt{perform}\ \mathscr{C}\ \langle k, \langle \tilde{e}, \tilde{v}\rangle\rangle)$ Inversion
      $e'_1 = \tilde{E}^*[\texttt{handle}_s(\mathtt{h}_{\mathscr{C}}, \texttt{perform}\ \mathscr{C}\ \langle k, \langle \tilde{e}, \tilde{v}\rangle\rangle)]$ Case 1 EQ
      $e'_1 \Longrightarrow^* \text{if } \tilde{e}\ \tilde{v} \text{ then } \left(\lambda x.\tilde{E}^*[x]\right)\ \tilde{v} \text{ else } \texttt{error}_k$
      $e_2 = E^*\left[\mathsf{check}^k_j(e\ v, v)\right]$

$e\ v \sim \tilde{e}\ \tilde{v}$

$\mathrm{EQ}(e_2, \text{if } \tilde{e}\ \tilde{v} \text{ then } (\lambda x.\tilde{E}^*[x])\ \tilde{v} \text{ else } \mathtt{error}_k)$ Case 2 EQ

$e_2' = \text{if } \tilde{e}\ \tilde{v} \text{ then } (\lambda x.\tilde{E}^*[x])\ \tilde{v} \text{ else } \mathtt{error}_k$ Choose

– Case $E^+ = E_1^+\left[E_1^*\left[\mathsf{check}_{j_1}^{k_1}(\square, v_1)\right]\right]$

$e_2 =$
$E_1^+\left[E_1^*\left[\mathsf{check}_{j_1}^{k_1}\left(E^*\left[\mathsf{check}_j^k(e\ v, v)\right], v_1\right)\right]\right] = E_2^+\left[E^*\left[\mathsf{check}_j^k(e\ v, v)\right]\right]$
$\tilde{E}_2^+ = \text{if } \square \text{ then } (\lambda x.\tilde{E}_1^+\left[\tilde{E}_1^*[x]\right])\ \tilde{v}_1 \text{ else } \mathtt{error}_{k_1}$
$E_2^+\left[E^*\left[\mathsf{check}_j^k(\square, v)\right]\right] \sim \text{if } \square \text{ then } (\lambda x.\tilde{E}_2^+\left[\tilde{E}^*[x]\right])\ \tilde{v} \text{ else } \mathtt{error}_k$
$e_1' = \text{if } \tilde{E}^*[\tilde{e}_3] \text{ then } (\lambda x.\tilde{E}_1^+\left[\tilde{E}_1^*[x]\right])\ \tilde{v}_1 \text{ else } \mathtt{error}_{k_1}$ Case 2 EQ
$\tilde{e}_3 = \mathtt{handle}_s(\mathtt{h}_{\mathscr{C}}, \mathtt{perform}\ \mathscr{C}\ \langle k, \langle \tilde{e}, \tilde{v}\rangle\rangle)$ Inversion
$e_1' \Longrightarrow^* \text{if } \tilde{e}\ \tilde{v} \text{ then } \left(\lambda x.\ \text{if } \tilde{E}^*[x] \text{ then } (\lambda y.\tilde{E}_1^+\left[\tilde{E}_1^*[y]\right])\ \tilde{v}_1 \text{ else } \mathtt{error}_{k_1}\right)\ \tilde{v} \text{ else } \mathtt{error}_k$
$= \text{if } \tilde{e}\ \tilde{v} \text{ then } (\lambda x.\tilde{E}_2^+\left[\tilde{E}^*[x]\right])\ \tilde{v} \text{ else } \mathtt{error}_k$
$\mathrm{EQ}(e_2, \text{if } \tilde{e}\ \tilde{v} \text{ then } (\lambda x.\tilde{E}_2^+\left[\tilde{E}^*[x]\right])\ \tilde{v} \text{ else } \mathtt{error}_k)$ Case 2 EQ
$e_2' = \text{if } \tilde{e}\ \tilde{v} \text{ then } (\lambda x.\tilde{E}_2^+\left[\tilde{E}^*[x]\right])\ \tilde{v} \text{ else } \mathtt{error}_k$ Choose

▸ Case $e_3 = \mathsf{check}_j^k(\mathsf{true}, v) \dashrightarrow v = e_4$

Due to the restriction of EQ, only EQ case 2 matches this case.

$e_1 = E\left[\mathsf{check}_j^k(\mathsf{true}, v)\right] = E^+\left[E^*\left[\mathsf{check}_j^k(\mathsf{true}, v)\right]\right]$ Given and Lemma 0.4
$e_2 = E^+[E^*[v]]$

– Case $E^+ = \square$

$e_1 = E^*\left[\mathsf{check}_j^k(\mathsf{true}, v)\right]$
$e_2 = E^*[v]$
$e_1' = \text{if true then } (\lambda x.\tilde{E}^*[x])\ \tilde{v} \text{ else } \mathtt{error}_k$ EQ case 2
$e_1' \Longrightarrow^* \tilde{E}^*[\tilde{v}]$
$E^*[v] \sim \tilde{E}^*[\tilde{v}]$ Lemma 0.1
$\mathrm{EQ}(e_2, \tilde{E}^*[\tilde{v}])$ EQ case 1
$e_2' = \tilde{E}^*[\tilde{v}]$ Choose

– Case $E^+ = E_1^+\left[E_1^*\left[\mathsf{check}_{j_1}^{k_1}(\square, v_1)\right]\right]$

$e_2 = E_1^+\left[E_1^*\left[\mathsf{check}_{j_1}^{k_1}(E^*[v], v_1)\right]\right]$
$e_1 = E_1^+\left[E_1^*\left[\mathsf{check}_{j_1}^{k_1}\left(E^*\left[\mathsf{check}_j^k(\mathsf{true}, v)\right], v_1\right)\right]\right] =$
$E_2^+\left[E^*\left[\mathsf{check}_j^k(\mathsf{true}, v)\right]\right]$
$e_1' = \text{if true then } (\lambda y.\tilde{E}_2^+\left[\tilde{E}^*[y]\right])\ \tilde{v} \text{ else } \mathtt{error}_k$ Case 2 EQ
$E_2^+ \sim \text{if } \square \text{ then } (\lambda x.\tilde{E}_1^+[E_1^*[x]])\ \tilde{v}_1 \text{ else } \mathtt{error}_{k_1} = \tilde{E}_2^+$
$e_1' \Longrightarrow^* \tilde{E}_2^+\left[\tilde{E}^*[\tilde{v}]\right]$
$E^*[v] \sim \tilde{E}^*[\tilde{v}]$ Lemma 0.1

$\mathrm{EQ}\big(e_2, \text{if } \tilde{E}^*[\tilde{v}] \text{ then } (\lambda x.\tilde{E}_1^+[E_1^*[x]])\ \tilde{v}_1 \text{ else } \mathtt{error}_{k_1}\big)$ Case 2 EQ

$\mathrm{EQ}\big(e_2, \tilde{E}_2^+\left[\tilde{E}^*[\tilde{v}]\right]\big)$ Replace with $\tilde{E}_2^+$

$e_2' = \tilde{E}_2^+\left[\tilde{E}^*[\tilde{v}]\right]$ Choose

- Case $e_3 = \mathsf{check}_j^k(\mathsf{false}, v) \dashrightarrow \mathsf{blame}_j^k = e_4$

$e_1 = E\left[\mathsf{check}_j^k(\mathsf{false}, v)\right] = E^+\left[E^*\left[\mathsf{check}_j^k(\mathsf{false}, v)\right]\right]$ Given and Lemma 0.4

$e_2 = E^+\left[E^*\left[\mathsf{blame}_j^k\right]\right]$

  - Case $E^+ = \square$

$e_1 = E^*\left[\mathsf{check}_j^k(\mathsf{false}, v)\right]$

$e_2 = E^*\left[\mathsf{blame}_j^k\right]$

$e_1' = \text{if false then } (\lambda x.\tilde{E}^*[x])\ \tilde{v} \text{ else } \mathtt{error}_k$ EQ case 2

$e_1' \Longrightarrow^* \mathtt{error}_k$

$\mathrm{EQ}(e_2, \mathtt{error}_k)$ Case 3 EQ

$e_2' = \mathtt{error}_k$ Choose

  - Case $E^+ = E_1^+\left[E_1^*\left[\mathsf{check}_{j_1}^{k_1}(\square, v_1)\right]\right]$

$e_2 = E_1^+\left[E_1^*\left[\mathsf{check}_{j_1}^{k_1}\left(E^*\left[\mathsf{blame}_j^k\right], v_1\right)\right]\right]$

$e_1 = E_1^+\left[E_1^*\left[\mathsf{check}_{j_1}^{k_1}\left(E^*\left[\mathsf{check}_j^k(\mathsf{false}, v)\right], v_1\right)\right]\right] = E_2^+\left[E^*\left[\mathsf{check}_j^k(\mathsf{false}, v)\right]\right]$

$e_1' = \text{if false then } (\lambda y.\tilde{E}_2^+\left[\tilde{E}^*[y]\right])\ \tilde{v} \text{ else } \mathtt{error}_k$ Case 2 EQ

$e_1' \Longrightarrow^* \mathtt{error}_k$

$\mathrm{EQ}(e_2, \mathtt{error}_k)$ Case 3 EQ

$e_2' = \mathtt{error}_k$ Choose

- Case $\mathsf{mon}_j^{k,l}(\kappa_1 \to \kappa_2, v) \dashrightarrow \mathsf{guard}_j^{k,l}(\kappa_1 \to \kappa_2, v)$

$e_1 = E^+\left[E^*\left[\mathsf{mon}_j^{k,l}(\kappa_1 \to \kappa_2, v)\right]\right]$ Lemma 0.4

  - Case $E^+ = \square$

$e_1 = E^*\left[\mathsf{mon}_j^{k,l}(\kappa_1 \to \kappa_2, v)\right]$

$e_2 = E^*\left[\mathsf{guard}_j^{k,l}(\kappa_1 \to \kappa_2, v)\right]$

$E^* \sim \tilde{E}^*$ Lemma 0.3

$e_3 \sim \tilde{e}_3$ Lemma 0.3

$\tilde{e}_3 = (\lambda f.\tilde{e}_{3a})\ \tilde{v}; e_{3a} = \lambda x.\mathsf{mon}_j^{k,l}\left(\kappa_2, f\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\right)$

$e_1 \sim \tilde{E}^*[\tilde{e}_3]$ Lemma 0.1

$e_1' = \tilde{E}^*[\tilde{e}_3]$

$e_1' \Longrightarrow^* \tilde{E}^*[\tilde{e}_{3a}[f := \tilde{v}]]$

$e_{3a}[f := v] = \lambda x.\mathsf{mon}_j^{k,l}\left(\kappa_2, v\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\right)$

$e_{3a}[f := v] \sim \tilde{e}_{3a}[f := \tilde{v}]$ Lemma 0.2

$\mathsf{guard}_j^{k,l}(\kappa_1 \to \kappa_2, v) \sim \tilde{e}_{3a}[f := \tilde{v}]$ — Relation
$e_2 \sim \tilde{E}^*[\tilde{e}_{3a}[f := \tilde{v}]]$ — Lemma 0.1
$\mathrm{EQ}(e_2, \tilde{E}^*[\tilde{e}_{3a}[f := \tilde{v}]])$ — Case 1 EQ
$e_2' = \tilde{E}^*[\tilde{e}_{3a}[f := \tilde{v}]]$ — Choose

– Case $E^+ = E_1^+\left[E_1^*\left[\mathsf{check}_{j_1}^{k_1}(\square, v_1)\right]\right]$

$e_1 = E_1^+\left[E_1^*\left[\mathsf{check}_{j_1}^{k_1}\left(E^*\left[\mathsf{mon}_j^{k,l}(\kappa_1 \to \kappa_2, v)\right], v_1\right)\right]\right]$
$e_2 = E_1^+\left[E_1^*\left[\mathsf{check}_{j_1}^{k_1}\left(E^*\left[\mathsf{guard}_j^{k,l}(\kappa_1 \to \kappa_2, v)\right], v_1\right)\right]\right]$
$E^* \sim \tilde{E}^*$ — Lemma 0.3
$e_3 \sim \tilde{e}_3$ — Lemma 0.3
$\tilde{e}_3 = (\lambda f.\tilde{e}_{3a})\ \tilde{v}; e_{3a} = \lambda x.\mathsf{mon}_j^{k,l}\left(\kappa_2, f\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\right)$
$e_1 \sim \tilde{E}^*[\tilde{e}_3]$ — Lemma 0.1
$e_1' = \text{if } \tilde{E}^*[\tilde{e}_3] \text{ then } \left(\lambda x.\tilde{E}_1^+\left[\tilde{E}_1^*[x]\right]\right)\ \tilde{v}_1 \text{ else } \mathtt{error}_{k_1}$ — Case 2 EQ
$e_1' \Longrightarrow^* \text{if } \tilde{E}^*[\tilde{e}_{3a}[f := \tilde{v}]] \text{ then } \left(\lambda x.\tilde{E}_1^+\left[\tilde{E}_1^*[x]\right]\right)\ \tilde{v}_1 \text{ else } \mathtt{error}_{k_1}$
$e_{3a}[f := v] = \lambda x.\mathsf{mon}_j^{k,l}\left(\kappa_2, v\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\right)$
$e_{3a}[f := v] \sim \tilde{e}_{3a}[f := \tilde{v}]$ — Lemma 0.2
$\mathsf{guard}_j^{k,l}(\kappa_1 \to \kappa_2, v) \sim \tilde{e}_{3a}[f := \tilde{v}]$ — Relation
$E^*\left[\mathsf{guard}_j^{k,l}(\kappa_1 \to \kappa_2, v)\right] \sim \tilde{E}^*[\tilde{e}_{3a}[f := \tilde{v}]]$ — Lemma 0.1
$\mathrm{EQ}\left(e_2, \text{if } \tilde{E}^*[\tilde{e}_{3a}[f := \tilde{v}]] \text{ then } \left(\lambda x.\tilde{E}_1^+\left[\tilde{E}_1^*[x]\right]\right)\ \tilde{v}_1 \text{ else } \mathtt{error}_{k_1}\right)$ — Case 2 EQ
$e_2' = \text{if } \tilde{E}^*[\tilde{e}_{3a}[f := \tilde{v}]] \text{ then } \left(\lambda x.\tilde{E}_1^+\left[\tilde{E}_1^*[x]\right]\right)\ \tilde{v}_1 \text{ else } \mathtt{error}_{k_1}$ — Choose

▸ Case $\mathsf{mon}_j^{k,l}(\kappa_1 \to_i \lambda y.\kappa_2, v) \dashrightarrow \mathsf{guard}_j^{k,l}(\kappa_1 \to_i \lambda y.\kappa_2, v)$

Similarly to the functional case. Replacement variable is a fresh f, it does not affect anything.

$e_1 = E^+\left[E^*\left[\mathsf{mon}_j^{k,l}(\kappa_1 \to_i \lambda y.\kappa_2, v)\right]\right]$ — Lemma 0.4

– Case $E^+ = \square$

$e_1 = E^*\left[\mathsf{mon}_j^{k,l}(\kappa_1 \to_i \lambda y.\kappa_2, v)\right]$
$e_2 = E^*\left[\mathsf{guard}_j^{k,l}(\kappa_1 \to_i \lambda y.\kappa_2, v)\right]$
$E^* \sim \tilde{E}^*$ — Lemma 0.3
$e_3 \sim \tilde{e}_3$ — Lemma 0.3
$\tilde{e}_3 = (\lambda f.\tilde{e}_{3a})\ \tilde{v}$
$e_{3a} = \lambda x.\mathsf{mon}_j^{k,l}\left(\kappa_3, f\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\right); \kappa_3 = \kappa_2\left[y := \mathsf{mon}_j^{l,j}(\kappa_1, x)\right]$
$e_1 \sim \tilde{E}^*[\tilde{e}_3]$ — Lemma 0.1
$e_1' = \tilde{E}^*[\tilde{e}_3]$
$e_1' \Longrightarrow^* \tilde{E}^*[\tilde{e}_{3a}[f := \tilde{v}]]$
$e_{3a}[f := v] = \lambda x.\mathsf{mon}_j^{k,l}\left(\kappa_3, v\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\right)$

$e_{3a}[f := v] \sim \tilde{e}_{3a}[f := \tilde{v}]$ Lemma 0.2
$\mathsf{guard}_j^{k,l}(\kappa_1 \to_i \lambda y.\kappa_2, v) \sim \tilde{e}_{3a}[f := \tilde{v}]$ Relation
$e_2 \sim \tilde{E}^*[\tilde{e}_{3a}[f := \tilde{v}]]$ Lemma 0.1
$\mathrm{EQ}\big(e_2, \tilde{E}^*[\tilde{e}_{3a}[f := \tilde{v}]]\big)$ Case 1 EQ
$e_2' = \tilde{E}^*[\tilde{e}_{3a}[f := \tilde{v}]]$ Choose

– Case $E^+ = E_1^+\left[E_1^*\left[\mathsf{check}_{j_1}^{k_1}(\square, v_1)\right]\right]$

$e_1 = E_1^+\left[E_1^*\left[\mathsf{check}_{j_1}^{k_1}\left(E^*\left[\mathsf{mon}_j^{k,l}(\kappa_1 \to_i \lambda y.\kappa_2, v)\right], v_1\right)\right]\right]$
$e_2 = E_1^+\left[E_1^*\left[\mathsf{check}_{j_1}^{k_1}\left(E^*\left[\mathsf{guard}_j^{k,l}(\kappa_1 \to_i \lambda y.\kappa_2, v)\right], v_1\right)\right]\right]$
$E^* \sim \tilde{E}^*$ Lemma 0.3
$e_3 \sim \tilde{e}_3$ Lemma 0.3
$\tilde{e}_3 = (\lambda f.\tilde{e}_{3a})\ \tilde{v}$
$e_{3a} = \lambda x.\mathsf{mon}_j^{k,l}\big(\kappa_3, f\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\big); \kappa_3 = \kappa_2\left[y := \mathsf{mon}_j^{l,j}(\kappa_1, x)\right]$
$e_1 \sim \tilde{E}^*[\tilde{e}_3]$ Lemma 0.1
$e_1' = \text{if } \tilde{E}^*[\tilde{e}_3] \text{ then } \left(\lambda x.\tilde{E}_1^+\left[\tilde{E}_1^*[x]\right]\right)\tilde{v}_1 \text{ else } \mathtt{error}_{k_1}$ Case 2 EQ
$e_1' \Longrightarrow^* \text{if } \tilde{E}^*[\tilde{e}_{3a}[f := \tilde{v}]] \text{ then } \left(\lambda x.\tilde{E}_1^+\left[\tilde{E}_1^*[x]\right]\right)\tilde{v}_1 \text{ else } \mathtt{error}_{k_1}$
$e_{3a}[f := v] = \lambda x.\mathsf{mon}_j^{k,l}\big(\kappa_3, v\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\big)$
$e_{3a}[f := v] \sim \tilde{e}_{3a}[f := \tilde{v}]$ Lemma 0.2
$\mathsf{guard}_j^{k,l}(\kappa_1 \to_i \lambda y.\kappa_2, v) \sim \tilde{e}_{3a}[f := \tilde{v}]$ Relation
$E^*\left[\mathsf{guard}_j^{k,l}(\kappa_1 \to_i \lambda y.\kappa_2, v)\right] \sim \tilde{E}^*[\tilde{e}_{3a}[f := \tilde{v}]]$ Lemma 0.1
$\mathrm{EQ}\big(e_2, \text{if } \tilde{E}^*[\tilde{e}_{3a}[f := \tilde{v}]] \text{ then } \left(\lambda x.\tilde{E}_1^+\left[\tilde{E}_1^*[x]\right]\right)\tilde{v}_1 \text{ else } \mathtt{error}_{k_1}\big)$ Case 2 EQ
$e_2' = \text{if } \tilde{E}^*[\tilde{e}_{3a}[f := \tilde{v}]] \text{ then } \left(\lambda x.\tilde{E}_1^+\left[\tilde{E}_1^*[x]\right]\right)\tilde{v}_1 \text{ else } \mathtt{error}_{k_1}$ Choose

▸ Case $e_3 = \mathsf{guard}_j^{k,l}(\kappa_1 \to \kappa_2, v_1)\ v_2 \dashrightarrow \mathsf{mon}_j^{k,l}\big(\kappa_2, v_1\ \mathsf{mon}_j^{l,k}(\kappa_1, v_2)\big) = e_4$

$e_1 = E^+\left[E^*\left[\mathsf{guard}_j^{k,l}(\kappa_1 \to \kappa_2, v_1)\ v_2\right]\right]$ Lemma 0.4

– Case $E^+ = \square$

$e_1 = E^*\left[\mathsf{guard}_j^{k,l}(\kappa_1 \to \kappa_2, v_1)\ v_2\right]$
$E^* \sim \tilde{E}^*$ Lemma 0.3
$e_3 \sim \tilde{e}_3$ Lemma 0.3
$\tilde{e}_3 = \lambda x.\tilde{e}_{3a}\ \tilde{v}_2; e_{3a} = \mathsf{mon}_j^{k,l}\big(\kappa_2, v_1\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\big)$
$e_1 \sim \tilde{E}^*[\tilde{e}_3]$ Lemma 0.1
$e_1' = \tilde{E}^*[\tilde{e}_3]$ Case 1 EQ
$e_1' \Longrightarrow^* \tilde{E}^*[\tilde{e}_{3a}[x := \tilde{v}_2]]$
$e_{3a}[x := v_2] = \mathsf{mon}_j^{k,l}\big(\kappa_2, v_1\ \mathsf{mon}_j^{l,k}(\kappa_1, v_2)\big)$
$e_{3a}[x := v_2] \sim \tilde{e}_{3a}[x := \tilde{v}_2]$ Lemma 0.2
$e_2 = E^*\left[\mathsf{mon}_j^{k,l}\big(\kappa_2, v_1\ \mathsf{mon}_j^{l,k}(\kappa_1, v_2)\big)\right] \sim \tilde{E}^*[\tilde{e}_{3a}[x := \tilde{v}_2]]$ Lemma 0.1

$\mathrm{EQ}\big(e_2, \tilde{E}^*[\tilde{e}_{3a}[x := \tilde{v}_2]]\big)$ Case 1 EQ

$e_2' = \tilde{E}^*[\tilde{e}_{3a}[x := \tilde{v}_2]]$ Choose

– Case $E^+ = E_1^+\left[E_1^*\left[\mathsf{check}_{j_1}^{k_1}(\square, v_3)\right]\right]$

$e_1 = E_1^+\left[E_1^*\left[\mathsf{check}_{j_1}^{k_1}\left(E^*\left[\mathsf{guard}_j^{k,l}(\kappa_1 \to \kappa_2, v_1)\ v_2\right], v_3\right)\right]\right]$

$E^* \sim \tilde{E}^*$ Lemma 0.3

$e_3 \sim \tilde{e}_3$ Lemma 0.3

$\tilde{e}_3 = \lambda x.\tilde{e}_{3a}\ \tilde{v}_2; e_{3a} = \mathsf{mon}_j^{k,l}\left(\kappa_2, v_1\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\right)$

$E^*[\tilde{e}_3] \sim \tilde{E}^*[\tilde{e}_3]$ Lemma 0.1

$e_1' = \text{if } \tilde{E}^*[\tilde{e}_3] \text{ then } \left(\lambda x.\tilde{E}_1^+\left[\tilde{E}^*[x]\right]\right)\ \tilde{v}_3 \text{ else } \mathtt{error}_{k_1}$ Case 2 EQ

$e_1' \Longrightarrow^* \text{if } \tilde{E}^*[\tilde{e}_{3a}[x := \tilde{v}_2]] \text{ then } \left(\lambda x.\tilde{E}_1^+\left[\tilde{E}^*[x]\right]\right)\ \tilde{v}_3 \text{ else } \mathtt{error}_{k_1}$

$e_{3a}[x := v_2] = \mathsf{mon}_j^{k,l}\left(\kappa_2, v_1\ \mathsf{mon}_j^{l,k}(\kappa_1, v_2)\right)$

$e_{3a}[x := v_2] \sim \tilde{e}_{3a}[x := \tilde{v}_2]$ Lemma 0.2

$E^*\left[\mathsf{mon}_j^{k,l}\left(\kappa_2, v_1\ \mathsf{mon}_j^{l,k}(\kappa_1, v_2)\right)\right] \sim \tilde{E}^*[\tilde{e}_{3a}[x := \tilde{v}_2]]$ Lemma 0.1

$e_2 = E_1^+\left[E_1^*\left[\mathsf{check}_{j_1}^{k_1}\left(E^*\left[\mathsf{mon}_j^{k,l}\left(\kappa_2, v_1\ \mathsf{mon}_j^{l,k}(\kappa_1, v_2)\right)\right], v_3\right)\right]\right]$

$\mathrm{EQ}\Big(e_2, \text{if } \tilde{E}^*[\tilde{e}_{3a}[x := \tilde{v}_2]] \text{ then } \left(\lambda x.\tilde{E}_1^+\left[\tilde{E}^*[x]\right]\right)\ \tilde{v}_3 \text{ else } \mathtt{error}_{k_1}\Big)$ Case 2 EQ

$e_2' = \text{if } \tilde{E}^*[\tilde{e}_{3a}[x := \tilde{v}_2]] \text{ then } \left(\lambda x.\tilde{E}_1^+\left[\tilde{E}^*[x]\right]\right)\ \tilde{v}_3 \text{ else } \mathtt{error}_{k_1}$ Choose

▸ Case $e_3 = \mathsf{guard}_j^{k,l}(\kappa_1 \to_i \lambda y.\kappa_2, v_1)\ v_2 \dashrightarrow \mathsf{mon}_j^{k,l}\left(\kappa_3, v_1\ \mathsf{mon}_j^{l,k}(\kappa_1, v_2)\right) = e_4$ where $\kappa_3 = \kappa_2\left[y := \mathsf{mon}_j^{l,j}(\kappa_1, v_2)\right]$

Similar to the functional case. The only problem is $\kappa_4 = \kappa_2\left[y := \mathsf{mon}_j^{l,j}(\kappa_1, x)\right]$ where x is the new argument of the function created for relation. We need to reason $\kappa_4[x := v_2] = \kappa_3$, which is obvious.

$e_1 = E^+\left[E^*\left[\mathsf{guard}_j^{k,l}(\kappa_1 \to_i \lambda y.\kappa_2, v_1)\ v_2\right]\right]$ Lemma 0.4

– Case $E^+ = \square$

$e_1 = E^*\left[\mathsf{guard}_j^{k,l}(\kappa_1 \to_i \lambda y.\kappa_2, v_1)\ v_2\right]$

$E^* \sim \tilde{E}^*$ Lemma 0.3

$e_3 \sim \tilde{e}_3$ Lemma 0.3

$\tilde{e}_3 = \lambda x.\tilde{e}_{3a}\ \tilde{v}_2$

$e_{3a} = \mathsf{mon}_j^{k,l}\left(\kappa_4, v_1\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\right); \kappa_4 = \kappa_2\left[y := \mathsf{mon}_j^{l,j}(\kappa_1, x)\right]$

$e_1 \sim \tilde{E}^*[\tilde{e}_3]$ Lemma 0.1

$e_1' = \tilde{E}^*[\tilde{e}_3]$ Case 1 EQ

$e_1' \Longrightarrow^* \tilde{E}^*[\tilde{e}_{3a}[x := \tilde{v}_2]]$

$e_{3a}[x := v_2] = \mathsf{mon}_j^{k,l}\left(\kappa_4[x := v_2], v_1\ \mathsf{mon}_j^{l,k}(\kappa_1, v_2)\right)$

$\kappa_4[x := v_2]$

$= \kappa_2\left[y := \mathsf{mon}_j^{l,j}(\kappa_1, x)\right][x := v_2] = \kappa_2\left[y := \mathsf{mon}_j^{l,j}(\kappa_1, v_2)\right]$ $x \notin \mathrm{fv}(\kappa_2, \kappa_1)$

$= \kappa_3$

$e_{3a}[x := v_2] = \mathsf{mon}_j^{k,l}\big(\kappa_3, v_1\ \mathsf{mon}_j^{l,k}(\kappa_1, v_2)\big)$

$e_{3a}[x := v_2] \sim \tilde{e}_{3a}[x := \tilde{v}_2]$ — Lemma 0.2

$e_2 = E^*\big[\mathsf{mon}_j^{k,l}\big(\kappa_3, v_1\ \mathsf{mon}_j^{l,k}(\kappa_1, v_2)\big)\big] \sim \tilde{E}^*[\tilde{e}_{3a}[x := \tilde{v}_2]]$ — Lemma 0.1

$\mathrm{EQ}\big(e_2, \tilde{E}^*[\tilde{e}_{3a}[x := \tilde{v}_2]]\big)$ — Case 1 EQ

$e_2' = \tilde{E}^*[\tilde{e}_{3a}[x := \tilde{v}_2]]$ — Choose

– Case $E^+ = E_1^+\big[E_1^*\big[\mathsf{check}_{j_1}^{k_1}(\square, v_3)\big]\big]$

$e_1 = E_1^+\big[E_1^*\big[\mathsf{check}_{j_1}^{k_1}\big(E^*\big[\mathsf{guard}_j^{k,l}(\kappa_1 \to_i \lambda y.\kappa_2, v_1)\ v_2\big], v_3\big)\big]\big]$

$E^* \sim \tilde{E}^*$ — Lemma 0.3

$e_3 \sim \tilde{e}_3$ — Lemma 0.3

$\tilde{e}_3 = \lambda x.\tilde{e}_{3a}\ \tilde{v}_2$

$e_{3a} = \mathsf{mon}_j^{k,l}\big(\kappa_4, v_1\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\big); \kappa_4 = \kappa_2\big[y := \mathsf{mon}_j^{l,j}(\kappa_1, x)\big]$

$E^*[\tilde{e}_3] \sim \tilde{E}^*[\tilde{e}_3]$ — Lemma 0.1

$e_1' = \text{if } \tilde{E}^*[\tilde{e}_3] \text{ then } \big(\lambda x.\tilde{E}_1^+\big[\tilde{E}^*[x]\big]\big)\ \tilde{v}_3 \text{ else } \mathtt{error}_{k_1}$ — Case 2 EQ

$e_1' \Longrightarrow^* \text{if } \tilde{E}^*[\tilde{e}_{3a}[x := \tilde{v}_2]] \text{ then } \big(\lambda x.\tilde{E}_1^+\big[\tilde{E}^*[x]\big]\big)\ \tilde{v}_3 \text{ else } \mathtt{error}_{k_1}$

$e_{3a}[x := v_2] = \mathsf{mon}_j^{k,l}\big(\kappa_4[x := v_2], v_1\ \mathsf{mon}_j^{l,k}(\kappa_1, v_2)\big)$

$\kappa_4[x := v_2]$

$= \kappa_2\big[y := \mathsf{mon}_j^{l,j}(\kappa_1, x)\big][x := v_2] = \kappa_2\big[y := \mathsf{mon}_j^{l,j}(\kappa_1, v_2)\big]$ — $x \notin \mathrm{fv}(\kappa_2, \kappa_1)$

$= \kappa_3$

$e_{3a}[x := v_2] = \mathsf{mon}_j^{k,l}\big(\kappa_3, v_1\ \mathsf{mon}_j^{l,k}(\kappa_1, v_2)\big)$

$e_{3a}[x := v_2] \sim \tilde{e}_{3a}[x := \tilde{v}_2]$ — Lemma 0.2

$E^*\big[\mathsf{mon}_j^{k,l}\big(\kappa_3, v_1\ \mathsf{mon}_j^{l,k}(\kappa_1, v_2)\big)\big] \sim \tilde{E}^*[\tilde{e}_{3a}[x := \tilde{v}_2]]$ — Lemma 0.1

$e_2 = E_1^+\big[E_1^*\big[\mathsf{check}_{j_1}^{k_1}\big(E^*\big[\mathsf{mon}_j^{k,l}\big(\kappa_3, v_1\ \mathsf{mon}_j^{l,k}(\kappa_1, v_2)\big)\big], v_3\big)\big]\big]$

$\mathrm{EQ}\big(e_2, \text{if } \tilde{E}^*[\tilde{e}_{3a}[x := \tilde{v}_2]] \text{ then } \big(\lambda x.\tilde{E}_1^+\big[\tilde{E}^*[x]\big]\big)\ \tilde{v}_3 \text{ else } \mathtt{error}_{k_1}\big)$ — Case 2 EQ

$e_2' = \text{if } \tilde{E}^*[\tilde{e}_{3a}[x := \tilde{v}_2]] \text{ then } \big(\lambda x.\tilde{E}_1^+\big[\tilde{E}^*[x]\big]\big)\ \tilde{v}_3 \text{ else } \mathtt{error}_{k_1}$ — Choose

▸ Other cases follow similarly.

□